

16.5 Curl and Divergence

1 Warm Up

Problem 1: Draw the vector field $F(x, y) = x\hat{i}$

Problem 2: Find the equation of the plane contain the point $(1, 0, 2)$ and the line $x = t$, $y = 2t - 1$, $z = t$

2 Motivation

We are now going to learn about some properties of vector fields known as curl and divergence. This will allow us to analyze and understand what a vector field is doing. One way to get us in the mindset of properties of vector fields, we can imagine a fluid on a vector field and the vectors at each point may describe the velocities of particles of fluid or flow at that point.

3 Definitions & Theorems

1. Divergence at a point is a measure of how much *flow* enters the neighborhood around the point compared to how much flow leaves the neighborhood around the point. If more flow is entering the neighborhood than leaving it, then divergence is negative. If more flow is leaving the neighborhood than entering it then divergence is positive. If the two are the same, divergence is zero. This point is called incompressible. If every point is incompressible, the vector field is said to be incompressible.
 - Divergence negative $\rightarrow$ point gathers flow
 - Divergence positive $\rightarrow$ flow leaves the point (divergent)
 - Divergence zero $\rightarrow$ point is incompressible
2. If every point has positive divergence (i.e. is divergent) our vector field is called divergent.
3. Mathematically to find divergence of a point (x, y, z) on a vector field $F(x, y, z) = P\hat{i} + Q\hat{j} + R\hat{k}$, we define “del” $\nabla = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}$, just like we did with the gradient. Then we have divergence as follows.

$$\text{div}F = \nabla \cdot F = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

4. The other measure we have is known as **curl**. This measures the rotation of a vector field in the neighborhood of a point. If we imagine a paddle sticking up in the vertical z direction, we can think of curl as telling how the paddle will spin.

- Curl positive $\rightarrow$ counterclockwise rotation
- Curl negative $\rightarrow$ clockwise rotation
- Curl zero $\rightarrow$ no rotation (irrotational)

5. Mathematically to find curl to a vector field with component equations P, Q, R we take the cross product.

$$\text{curl}F = \nabla \times F = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \hat{i} - \left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z} \right) \hat{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \hat{k}$$

6. If we want to know the maximum rotation of our curl vectors, we can take the magnitude of our vector. This rotation would happen when our “paddle” lies parallel to our curl vector.
7. To keep things straight, it is important to know what the del operator is being applied to and what it produced.

	Applied to	Produces
Gradient	surface $\rightarrow \nabla f(x, y)$	(conservative) vector field
Divergence	vector field $\rightarrow \nabla \cdot F(x, y)$	(scalar field) surface
Curl	vector field $\rightarrow \nabla \times F(x, y)$	vector field

8. We get a useful theorem about conservative vector fields.

Theorem 1. *If F is a conservative vector field then $\text{curl}F = 0$ everywhere.*

This can be easily proved using Clairaut’s theorem.

4 Examples

Problem 1. Compare the curl and divergence to the vector fields $F(x, y, z) = x\hat{i}$ and $F(x, y, z) = x\hat{j}$.

Problem 2. Compare the curl and divergence to the vector fields $F(x, y, z) = x^2\hat{i} + y^2\hat{j}$ and $F(x, y, z) = y^2\hat{i} + x^2\hat{j}$.

Problem 3. Find the divergence and curl of $y^2\hat{i} + x^2\hat{j}$.

Problem 4. Find the divergence and curl of $F(x, y) = \frac{x}{\sqrt{x^2+y^2}}\hat{i} + \frac{y}{\sqrt{x^2+y^2}}\hat{j}$

Problem 5. Find the divergence and curl of $yz\hat{i} + xz\hat{j} + xy\hat{k}$.

Problem 6. Find the divergence and curl of $xy\hat{i} + xz\hat{j} + xyz^2\hat{k}$.

5 Scratch Work